Final Project Proposal

1.Certificate Name:

Asmaa Ahmed Hussein

2: Project title and description

Title: Online shoppers purchasing intention and behavior profiling Api

This dataset is used to build a classifier that predicts whether a user will complete a purchase transaction is highly for ecommerce conversation rate optimization . This project implements a comprehensive machine learning classification pipeline to predict a visitor's purchasing intent(Revenue: TRUE OR FALSE) using their behavioral session data. To optimize predictions, an unsupervised clustering .

3: Problem type

classification (supervised) coupled with clustering(unsupervised feature engineering.

the ultimate target is a binary classification output Revenue true or false .

4: Dataset

source : online shoppers purchasing intention dataset hosted on Kaggle(
Online Shoppers Purchase Intention)

size: 12,330 rows and 18 attribute .

target: Revenue (binary true if a purchase was made , otherwise false) .

Key Columns:

- Administrative, Informational, Product Related: Number of pages visited in each category.
- Duration: Time spent in each category.
- Bounce Rates, Exit Rates: Metrics measuring user engagement and site departure.
- Page Values: The average value of the page viewed before the user completed a transaction.
- Month, Region, Browser: Categorical data about the user.
- Revenue: The target label (TRUE = Purchase, FALSE = No Purchase).

Preprocessing plan: apply one-hot encoding to categorical session attributes normalize ranges with standardscaler and minmaxscaler, and execute an 80/20 train/test split.

5: Algorithms i plan to train

the architecture utilizes a combination of advanced clustering and multiple distinct classification algorithms.

- K-means used unsupervised feature engineering step to label users into browsing behavior profiles before classification.
- Gaussian Mixture Model(GMM) Evaluated against k-means to identify if soft-clustering probability features enhance downstream classifier score.
- Logistic Regression the baseline classification model used for benchmark metrics.

- Random Forest Classifier a rebuts ensemble method to interpret complex non linear combination of site interactions.
- Gradient Boosting classifier a sequential boosting technique used to minimize bias and boost accuracy on the tabular data.
- XGboosting classifier: an independent research algorithm built to extract maximum predictive precision from multi-feature retail environments.

6: Evaluation plan

Metrics for all models: Accuracy, Precision, Recall, and F1-Score

Best-model rule: Models will be ranked and selected based on the highest F1-Score on the validation split

he F1-score balances True Approvals vs False alarms, which is crucial since buying customers are typically a minority class in e-commerce traffic datasets

7. Deployment Sketch

Framework: FastAPi

Endpoint: `POST /predict`

input json example:

json

{

"Administrative: 54.2,

"Duration": 0.0,

"Product Related": 650.5,

"Bounce Rates": 0.0,

"Exit Rates": 0.02,

"Page Values": 34.8,

"Month": "May",

"Revenue": "True or False"

}

output json example:

json

{

"purchase_intention": "True",

"probability": 0.89,

"user_behavior_cluster": 2

}

6: Repository plan

```
online-shopper-intent-api/
├── dataset/
│   └── shopping.csv
├── src/
│   ├── preprocess.py # Session clustering and feature engineering
│   └── train.py # Training pipeline for Logistic, RF, Gradient Boosting, XGBoost
├── api/
│   └── app.py # FastAPI framework serving predictions
├── models/
│   ├── best_classifier.pkl
│   ├── behavior_clusterer.pkl
│   └── scaler.pkl
├── README.md # Installation and local deployment guide
├── requirements.txt # Project constraints
└── project_paper.md
```